

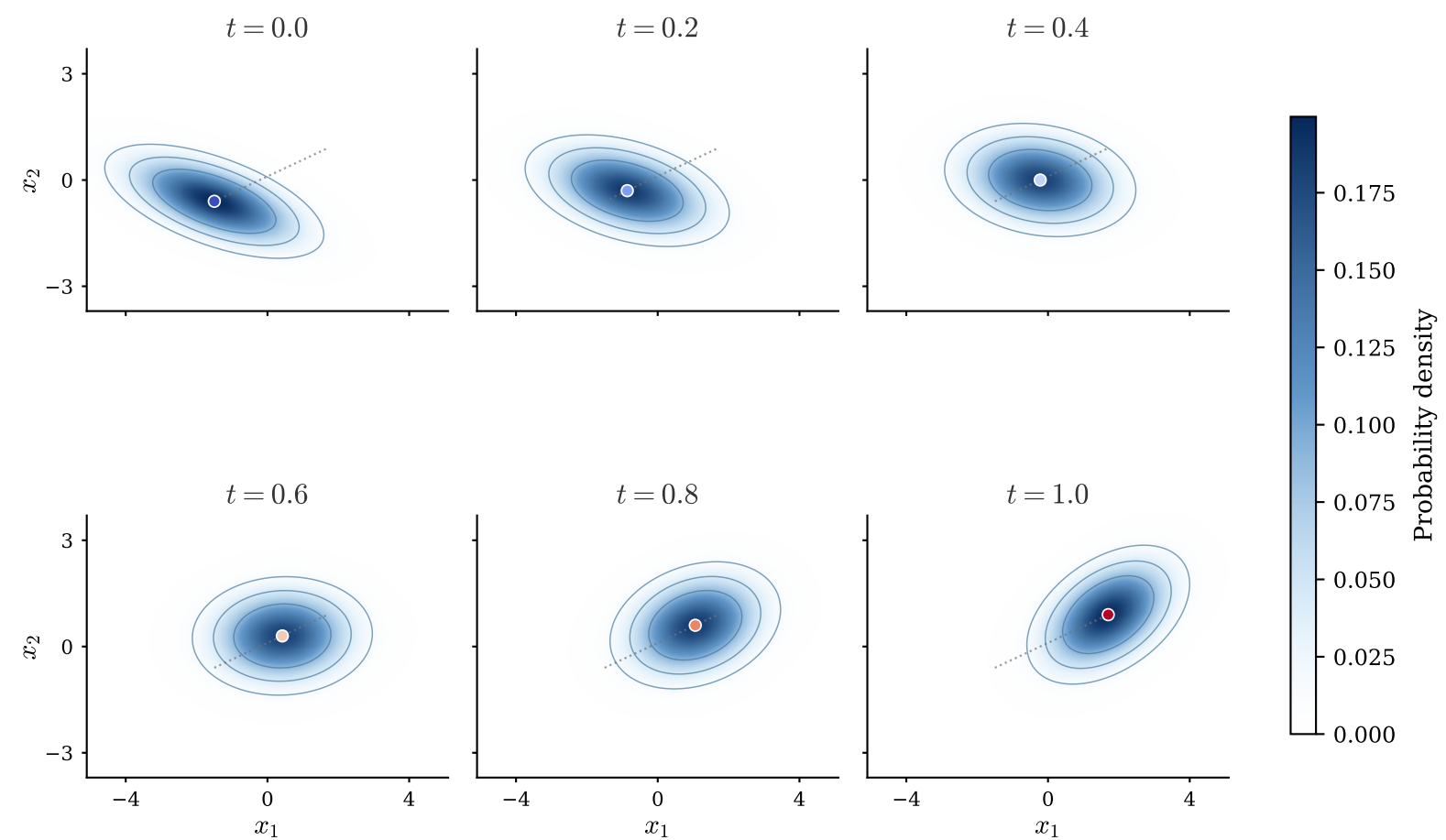
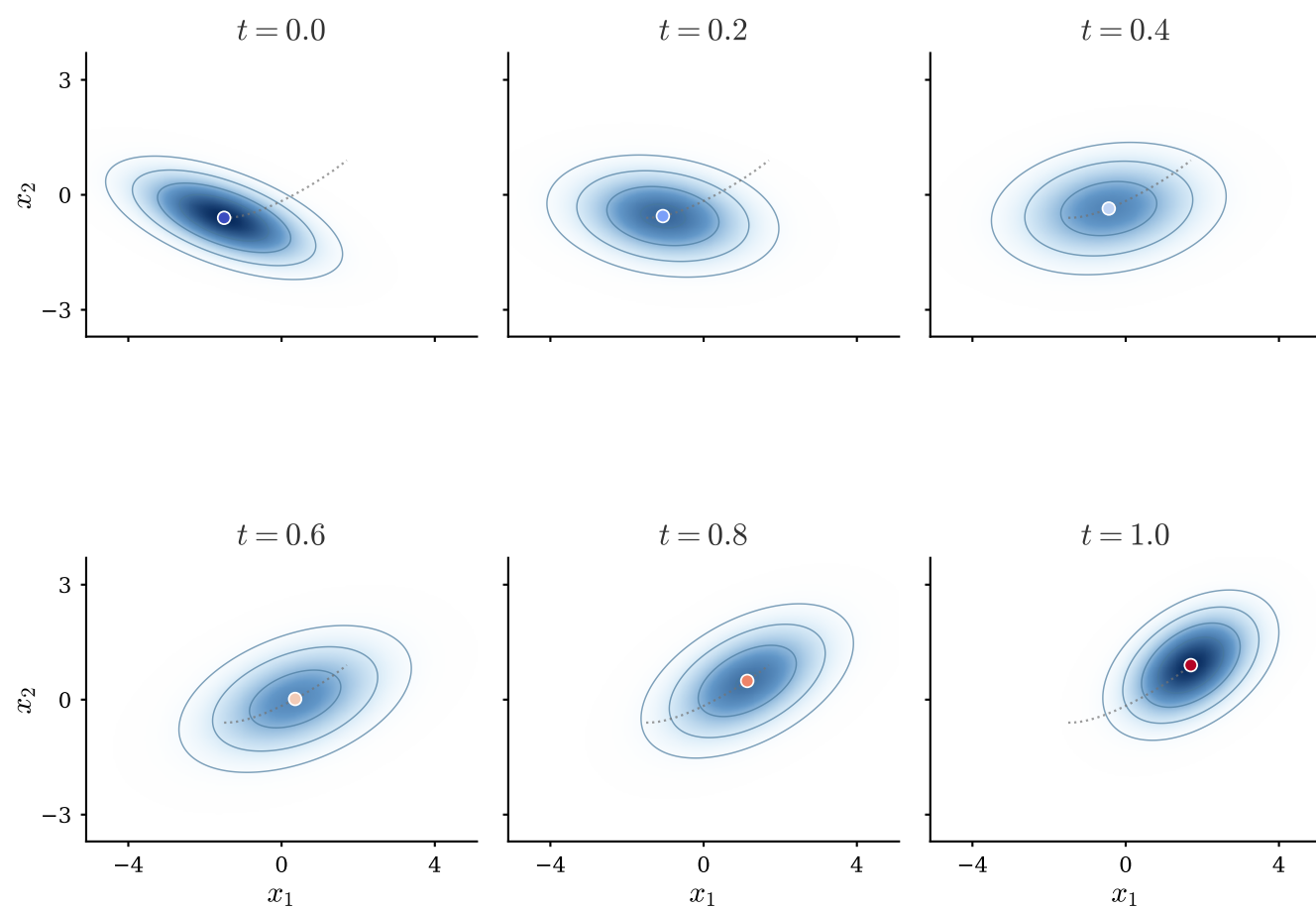
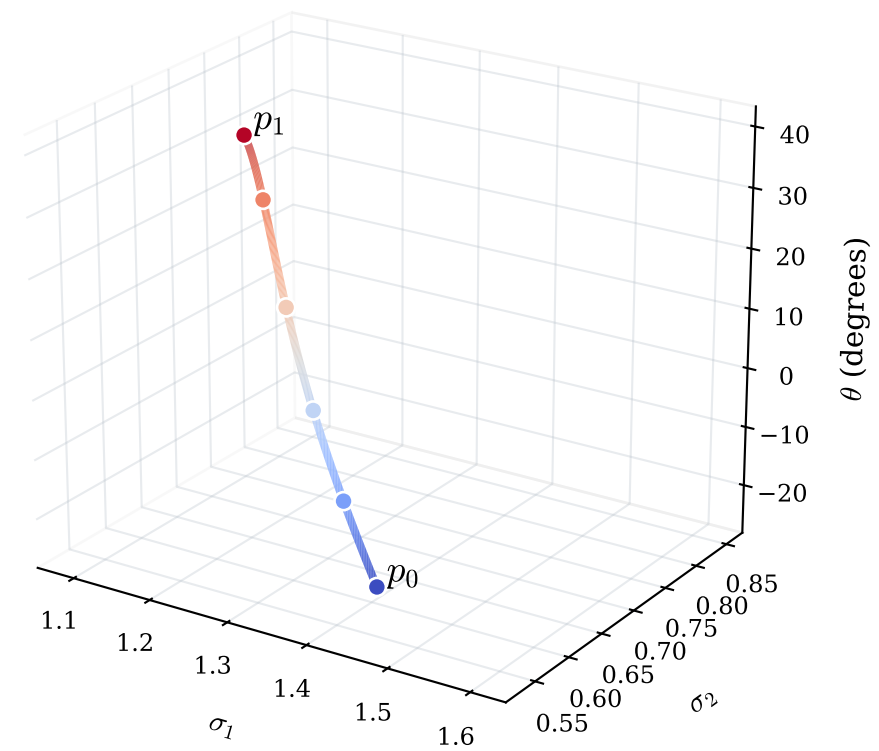
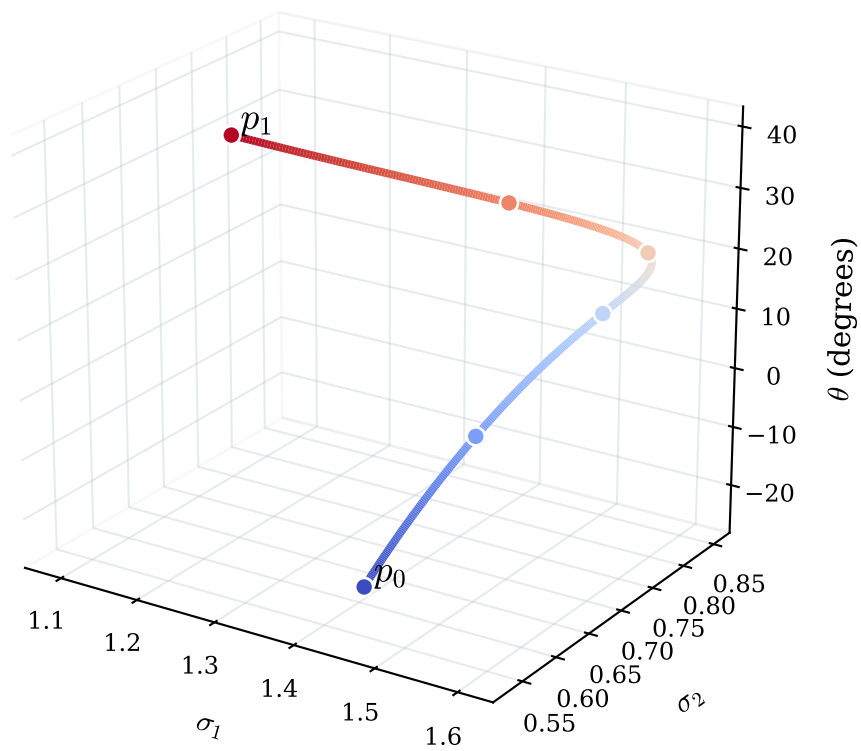
Two Gaussian geodesics: covariance shape and density evolution

$$\Sigma = R(\theta) \text{diag}(\sigma_1^2, \sigma_2^2) R(\theta)^T, \quad \sigma_1 > \sigma_2 > 0$$

Covariance projections of full Gaussian geodesics; means are included in the density snapshots.

Fisher-Rao — numerical Gaussian geodesic

Wasserstein-Otto — Gaussian geodesic



Dots and curve colors indicate geodesic time. Dotted traces in the snapshots show the mean paths.

Fisher-Rao couples mean and covariance; the Wasserstein-Otto mean moves on a straight segment.

Shared axes and density scale. Both covariance paths keep distinct eigenvalues and a continuous principal-axis angle.